

Neuro-Symbolic CSFM: Condition Dependent Source Flow Matching for Multi Objective Logic Synthesis

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Abstract

Multi objective logic synthesis is NP-hard combinatorial problem. Standard industrial compilers rely on greedy algebraic heuristics that frequently trap in local minima on high dimensional circuit manifolds. While deep generative models offer global structural exploration, they struggle to guarantee the strict Boolean equivalence required for viable hardware design. We introduce Neuro-Symbolic CSFM, a hybrid framework uniting continuous Conditional Source Flow Matching with discrete, equivalence preserving logic transformations. By conditioning a neural vector field on the initial circuit topology and a continuous Pareto preference vector, we model logic synthesis as a smooth, steerable trajectory in a latent representation space. The integrated target embedding serves as a neural heuristic to guide a discrete beam search. Because mapping complex input graphs to deterministic synthesis targets inherently risks latent space mode collapse, we introduce a topologically exact graph encoder combined with explicit variance and covariance regularizations to stabilize the continuous flow. Empirical evaluations on the EPFL benchmark suite demonstrate scalability: on an unseen 57,000 node divider circuit, CSFM escapes local minima to achieve a 63.6% reduction in area and delay Product, dynamically navigating the continuous Pareto frontier while guaranteeing 100% Boolean equivalence. **Code:** https://github.com/VladimerKhasia/neuro_symbolic_csfm

1. Introduction

Combinational logic synthesis is a cornerstone of Electronic Design Automation (EDA), tasked with minimizing the physical area and critical path delay of Boolean cir-

cuits. Modern industrial compilers, such as ABC (Brayton & Mishchenko, 2010), represent circuits as And-Inverter Graphs (AIGs) and optimize them through libraries of atomic, equivalence preserving algebraic transformations (e.g., rewriting and balancing). Because these transformations are not commutative and the discrete state space grows exponentially with circuit depth, standard heuristics act greedily. Consequently, they frequently fail to discover global optima on large scale logic manifolds.

Recent advancements in representation learning for global structural optimization including flow matching show promising results (Kim et al., 2026; Lipman et al., 2023; Liu et al., 2022; Cheng & Schwing, 2025; Chen et al., 2025). However, generating circuits directly with neural networks fails in practice because probabilistic models cannot guarantee exact Boolean equivalence, where even a single misplaced gate corrupts logic functionality. For this reason, machine learning methods restrict exploration to verified equivalence preserving transformations, applying rounding operations to continuous networks or use other workarounds (Haaswijk et al., 2018; Hosny et al., 2020; Hoogetboom et al., 2019; Tran et al., 2019).

To bridge the gap we build on previous research (Khasia, 2026; Watanabe, 1960; Barlow, 1989; Chen et al., 2018; Locatello et al., 2019; Yeredor, 2011; Jang et al., 2017; Maddison et al., 2017; van den Oord et al., 2017; Toffoli, 1980; Shende et al., 2003; Matsui, 1994; Gohr, 2019) and propose **Neuro-Symbolic CSFM**. Our framework formulates multi objective logic synthesis as a condition dependent normalizing flow in a latent representation space. Rather than generating graph edges directly, the neural flow integrates a continuous trajectory conditioned on user specified area and delay Pareto preferences. The terminal state of this trajectory acts as a highly accurate geometric heuristic, guiding a discrete, logically verified beam search.

To make this flow mathematically stable and physically grounded, we introduce two critical architectural priors. First, we design a topologically exact graph encoder that utilizes Kahn’s algorithm to explicitly extract critical path depth, ensuring the network is physically aware of logic delay rather than relying on shallow local message passing. Second, we introduce a suite of variance and covariance

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regularizations that strictly prevent latent space collapse, ensuring the continuous vector field provides robust, globally aware guidance across diverse circuit topologies.

In summary, this work makes the following primary contributions:

1. **Neuro-Symbolic Flow Matching Formulation:** We present a hybrid synthesis engine that bridges the continuous, iterative global targeting of flow matching with the strict Boolean equivalence guarantees of discrete heuristic beam search.
2. **Topological Encoding and Latent Regularization:** We introduce a depth aware graph encoder combined with variance and batch wise covariance regularizations that empirically eliminate representation mode collapse in deterministic optimization spaces.
3. **Dynamic Pareto Steering and Scalability:** We demonstrate that continuous preference conditioning accurately steers the discrete synthesis trajectory across the multi-objective frontier. Empirically, the framework scales to industrial complexities, escaping local minima to achieve a 63.6% area and delay Product reduction on a 57k node architecture.

2. Methodology

We introduce Neuro-Symbolic CSFM, a framework that unites continuous generative flow matching with discrete logic transformations. By conditioning on both the initial circuit structure and a continuous user preference weight, we cast multi objective logic synthesis as a smooth trajectory in a latent representation space. The endpoint of this continuous flow acts as a neural heuristic that guides a deduplicated discrete beam search, guaranteeing that all final transformations strictly preserve Boolean equivalence.

2.1. Symbolic Mapping and Continuous Formulation

We formalize the transformation of the discrete synthesis domain into continuous representation spaces as follows:

- $\mathcal{G} = (\mathcal{V}, \mathcal{E})$: An And-Inverter Graph (AIG) representing a combinational Boolean circuit.
- $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 5}$: Node feature matrix embedding categorical gate types (Primary Input, AND, NOT, Primary Output) concatenated with a normalized continuous topological depth.
- $\mathbf{w} = [\alpha_{\text{area}}, \alpha_{\text{delay}}]^\top \in \Delta^1$: Tradeoff preference vector residing on the unit simplex $\Delta^1 = \{\mathbf{w} \in \mathbb{R}_{\geq 0}^2 : \|\mathbf{w}\|_1 = 1\}$.
- $\mathbf{C} \in \mathbb{R}^d$: Latent embedding of the initial circuit graph with dimension $d = 128$.
- $\mathbf{X}_1 \in \mathbb{R}^d$: Target latent embedding, derived deterministically via domain specific boundary conditions.
- $\mathbf{X}_0 \sim p_\phi(\mathbf{X}_0 \mid \mathbf{C}, \mathbf{w})$: Source distribution conditioned on the initial circuit and preference vector.
- $v_\theta(\mathbf{X}_t, \mathbf{C}, \mathbf{w}, t)$: Neural vector field parameterized by weights θ at integration time $t \in [0, 1]$.
- ψ, ϕ, θ : Trainable neural network parameters representing the graph encoder, source generator, and velocity vector field, respectively.
- $\mathcal{A}$: The discrete action library of atomic structural transformations that preserve logical equivalence.

2.2. Topological Dual Pooling Graph Encoder

Standard graph neural network aggregations implicitly compress local connectivity but lack the global receptive field required to analyze long range critical path delays, which frequently span hundreds of topological levels in industrial circuits. To resolve this without incurring extreme computational depth, our parsing engine dynamically computes exact topological levels via Kahn’s algorithm in $\mathcal{O}(|\mathcal{V}| + |\mathcal{E}|)$ time. This normalized depth is concatenated with categorical gate identities—explicitly differentiating inversion edges from logical conjunctions—to form the initial node feature matrix $\mathbf{H}^{(0)} = \mathbf{X}$.

We process the graph via a three layer GraphSAGE architecture (Hamilton et al., 2017). The node update at layer $\ell \in \{1, 2, 3\}$ with hidden dimension $d_h = 64$ is defined as:

$$\mathbf{H}_i^{(\ell)} = \text{SAGE}_\ell \left(\mathbf{H}_i^{(\ell-1)}, \left\{ \mathbf{H}_j^{(\ell-1)} : j \in \mathcal{N}(i) \right\} \right). \quad (1)$$

To capture both the global node volume (proxy for area) and extreme structural pathways (proxy for delay), we decouple the graph into mean and maximum global pools:

$$\mathbf{h}_{\text{mean}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{H}_i^{(3)}, \quad [\mathbf{h}_{\text{max}}]_k = \max_{i \in \mathcal{V}} [\mathbf{H}_i^{(3)}]_k, \quad (2)$$

where the maximum is taken component wise. The final continuous latent representation is obtained via the affine projection $\mathbf{C} = \mathbf{W}_{\text{proj}}[\mathbf{h}_{\text{mean}} \parallel \mathbf{h}_{\text{max}}] + \mathbf{b}_{\text{proj}}$.

2.3. Conditional Flow Matching and Latent Regularization

During training, objective specific boundary targets $\mathbf{X}_1$ are gathered by aggressively applying deterministic heuristics (e.g., intensive area or delay mapping scripts) to the unoptimized circuit. However, directly regressing a single step

multilayer perceptron $\mathbf{C} \rightarrow \mathbf{X}_1$ frequently results in representation blurring in high dimensional graph spaces. We therefore model the translation as a Continuous Normalizing Flow, allowing iterative neural refinement toward the target.

To enable smooth multi objective steering, we train the network by sampling continuous Pareto preference vectors $\mathbf{w} \sim \text{Dirichlet}([1.0, 1.0])$. The conditional source generator outputs distribution parameters via a multilayer perceptron MLP $_{\phi}$:

$$[\boldsymbol{\mu}_{\phi}, \log \boldsymbol{\sigma}_{\phi}^2] = \text{MLP}_{\phi}([\mathbf{C} \parallel \mathbf{w}]), \quad (3)$$

$$\mathbf{X}_0 = \boldsymbol{\mu}_{\phi} + \exp\left(\frac{1}{2} \log \boldsymbol{\sigma}_{\phi}^2\right) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \quad (4)$$

Let $\mathbf{X}_t = (1 - t)\mathbf{X}_0 + t\mathbf{X}_1$ define the linear interpolation path, and $\Delta = \mathbf{X}_1 - \mathbf{X}_0$ denote the analytical velocity. The vector field v_{θ} is trained using an unnormalized expected squared error over the mini batch $\mathcal{B}$ to guarantee consistent dimensionality scaling:

$$\mathcal{L}_{\text{FM}}(\theta, \phi) = \mathbb{E}_{\mathcal{B}} \left[\sum_{k=1}^d ([v_{\theta}(\mathbf{X}_t, \mathbf{C}, \mathbf{w}, t)]_k - [\Delta]_k)^2 \right]. \quad (5)$$

Preventing Representation Collapse: A well known failure mode in deep metric learning without contrastive bounds is latent space collapse, where the graph encoder maps all topologies to a trivial point, falsely driving $\mathcal{L}_{\text{FM}}$ to zero. We inject three explicit structural priors to prevent this phenomenon. First, we constrain the variance of the generated source distribution:

$$\mathcal{L}_{\text{VarReg}}(\phi) = \frac{1}{2d} \mathbb{E}_{\mathcal{B}} \left[\sum_{k=1}^d (\sigma_{\phi,k}^2 - 1.0 - \log \sigma_{\phi,k}^2) \right]. \quad (6)$$

Second, we encourage directional alignment between the generated source and the deterministic target using cosine similarity:

$$\mathcal{L}_{\text{align}}(\phi) = \mathbb{E}_{\mathcal{B}} \left[1 - \frac{\mathbf{X}_0 \cdot \mathbf{X}_1}{\|\mathbf{X}_0\|_2 \|\mathbf{X}_1\|_2} \right]. \quad (7)$$

Finally, drawing inspiration from Variance Invariance Covariance Regularization (VICReg), we penalize the collapse of the encoded circuit embeddings $\mathbf{C}$ within a mini batch by forcing the standard deviation across the batch dimension to remain strictly active. Because $\mathbf{C}$ is produced by the graph encoder, this loss specifically regularizes the encoder parameters ψ :

$$\mathcal{L}_{\text{std}}(\psi) = \frac{1}{d} \sum_{k=1}^d \text{ReLU} \left(1.0 - \sqrt{\text{Var}_{\mathcal{B}}([\mathbf{C}]_k) + \epsilon} \right). \quad (8)$$

Because the architecture is trained jointly, gradients propagate through the continuous integration path into the latent

representations. The total joint objective optimized over all parameters (ψ, ϕ, θ) is scalarized as:

$$\mathcal{L}_{\text{total}}(\psi, \phi, \theta) = \mathcal{L}_{\text{FM}} + \lambda_{\text{var}} \mathcal{L}_{\text{VarReg}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{std}} \mathcal{L}_{\text{std}}. \quad (9)$$

The total joint objective is optimized as $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FM}} + \lambda_{\text{var}} \mathcal{L}_{\text{VarReg}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{std}} \mathcal{L}_{\text{std}}$.

2.4. Neuro-Symbolic Heuristic Beam Search

While the continuous flow produces highly accurate representations of the optimized geometry, directly decoding $\mathbf{Z}_{\text{target}}$ into an AIG structure risks introducing Boolean logical faults. We therefore utilize the target as a geometric guide for a strictly equivalence preserving heuristic beam search.

For an unseen state s_{init} , we set the origin to the deterministic mode $\mathbf{X}_{(0)} = \boldsymbol{\mu}_{\phi}([\mathbf{C} \parallel \mathbf{w}])$. The continuous target $\mathbf{Z}_{\text{target}}$ is derived by forward Euler integration over $K = 50$ discrete intervals ($\Delta t = 1/K$):

$$\mathbf{X}_{(k+1)} = \mathbf{X}_{(k)} + v_{\theta} \left(\mathbf{X}_{(k)}, \mathbf{C}, \mathbf{w}, \frac{k}{K} \right) \Delta t, \quad (10)$$

yielding $\mathbf{Z}_{\text{target}} = \mathbf{X}_{(K)}$.

The discrete search explores transformations using the scalarized evaluation function $f(s) = g(s) + h(s)$. The objective cost $g(s)$ computes the localized Pareto improvement relative to the starting circuit:

$$g(s) = \alpha_{\text{area}} \left(\frac{\text{Nodes}(s)}{\text{Nodes}(s_{\text{init}})} \right) + \alpha_{\text{delay}} \left(\frac{\text{Levels}(s)}{\text{Levels}(s_{\text{init}})} \right). \quad (11)$$

To prevent uncalibrated Euclidean distances from dominating the bounded Pareto cost, our neural heuristic computes a normalized cosine distance between the candidate embedding $\mathbf{Z}_{\text{curr}}$ and the integrated flow target:

$$h(s) = \gamma \left(1 - \frac{\mathbf{Z}_{\text{curr}} \cdot \mathbf{Z}_{\text{target}}}{\|\mathbf{Z}_{\text{curr}}\|_2 \|\mathbf{Z}_{\text{target}}\|_2} \right). \quad (12)$$

Because logic synthesis transformations frequently invert one another (e.g., cyclic rewrites), the search explicitly maintains a structural hash closed list to prune cyclical states and prevent beam degeneration (Algorithm 1).

2.5. Complexity Analysis

The computational demands of the system cleanly divide into continuous neural inference and discrete structural optimization. Let $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the maximum count of vertices and edges across evaluated circuit states.

Time Complexity: Graph feature extraction utilizing Kahn’s algorithm executes in $\mathcal{O}(|\mathcal{V}| + |\mathcal{E}|)$. The three layer GraphSAGE model evaluates neighborhood messages

Algorithm 1 NS CSFM

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1: Input: Initial circuit state  $s_{\text{init}}$ , preference  $\mathbf{w} \in \Delta^1$ .
2: Initialize: Search depth  $D_{\text{max}}$ , beam width  $B$ ,  $\gamma$ .
3:  $\mathbf{Z}_{\text{target}} \leftarrow \text{EulerRollout}(\text{DualPoolEncoder}(s_{\text{init}}), \mathbf{w}, K)$ 
4:  $\mathcal{B} \leftarrow \{s_{\text{init}}\}$ ,  $s_{\text{best}} \leftarrow s_{\text{init}}$ ,  $g_{\text{best}} \leftarrow 1.0$ 
5:  $\mathcal{V}_{\text{visited}} \leftarrow \{\text{StructuralHash}(s_{\text{init}})\}$ 
6: for  $d = 1$  to  $D_{\text{max}}$  do
7:    $\mathcal{C}_{\text{cand}} \leftarrow \emptyset$ 
8:   for each state  $s \in \mathcal{B}$  do
9:     for each valid transformation  $a \in \mathcal{A}$  do
10:       $s' \leftarrow \text{ApplyAction}(s, a)$ 
11:       $\text{hash} \leftarrow \text{StructuralHash}(s')$ 
12:      if  $\text{hash} \notin \mathcal{V}_{\text{visited}}$  then
13:         $\mathcal{V}_{\text{visited}} \leftarrow \mathcal{V}_{\text{visited}} \cup \{\text{hash}\}$ 
14:        Compute  $g(s')$  and bounded heuristic  $h(s')$ 
15:         $\mathcal{C}_{\text{cand}} \leftarrow \mathcal{C}_{\text{cand}} \cup \{(g(s') + h(s'), g(s'), s')\}$ 
16:        if  $g(s') < g_{\text{best}}$  then
17:           $g_{\text{best}} \leftarrow g(s')$ ;  $s_{\text{best}} \leftarrow s'$ 
18:        end if
19:      end if
20:    end for
21:  end for
22:  Sort  $\mathcal{C}_{\text{cand}}$  ascendingly by  $f(s) = g(s) + h(s)$ 
23:   $\mathcal{B} \leftarrow \text{Top } B \text{ states from } \mathcal{C}_{\text{cand}}$ 
24: end for
25: return  $s_{\text{best}}$ 

```

with asymptotic cost $\mathcal{O}(L(|\mathcal{V}| + |\mathcal{E}|)d_h)$, where $L = 3$ and $d_h = 64$. Continuous integration across $K = 50$ steps evaluates the velocity network with a cost of $\mathcal{O}(Kd^2)$, where $d = 128$. During the discrete search, the algorithm expands a deterministic maximum of $B \cdot D_{\text{max}} \cdot |\mathcal{A}|$ candidates. Because atomic structural transformations in logic synthesis act on localized functional cones, they scale proportionally to graph size. The overall time complexity is therefore bounded asymptotically by:

$$\mathcal{O}(B \cdot D_{\text{max}} \cdot |\mathcal{A}|((|\mathcal{V}| + |\mathcal{E}|)d_h) + Kd^2). \quad (13)$$

Because the execution scales linearly with circuit size rather than exponentially, runtime remains practical even for large industrial benchmarks.

Space Complexity: The neural networks contain approximately 5×10^5 parameters, requiring negligible persistent memory. Active memory consumption during inference is governed by the retained graph representations for the active beam $\mathcal{B}$ and the structural closed list $\mathcal{V}_{\text{visited}}$. Storing string hashes for visited states requires $\mathcal{O}(N_{\text{visited}})$, where $N_{\text{visited}} \leq B \cdot D_{\text{max}} \cdot |\mathcal{A}|$. Disk operations generated by the synthesis engine are pruned immediately upon evaluation, strictly limiting temporary storage overhead to $\mathcal{O}(B \cdot |\mathcal{A}| \cdot (|\mathcal{V}| + |\mathcal{E}|))$.

3. Experiments

We evaluate Neuro-Symbolic CSFM on the EPFL combinational benchmark suite. Our experiments are designed to empirically answer three primary questions: (1) Does continuous flow matching successfully guide discrete synthesis to outperform unguided (blind) heuristic search on large-scale circuits? (2) Do the proposed structural flow regularizations prevent latent mode collapse? (3) Can the continuous preference vector dynamically steer the discrete search across the Pareto multiobjective frontier?

3.1. Experimental Setup

Dataset and Training: The framework is trained on a subset of 15 EPFL arithmetic and random control circuits and tested on unseen benchmarks to strictly evaluate structural generalization. To model continuous multi objective trade-offs, we augment the dataset by sampling 4 continuous Pareto preference vectors $\mathbf{w} \sim \text{Dirichlet}([1.0, 1.0])$ per circuit. The neural ODE is trained for 50 epochs using a batch size of 32 to ensure gradient stability while preventing structural overfitting.

Search Configuration: For all experiments, we execute the heuristic beam search with a maximum depth of $D_{\text{max}} = 10$ and a beam width of $B = 4$. For neural guided runs, the heuristic scalarization weight is set to $\gamma = 2.0$. Equivalence is formally verified for all optimized circuits using ABC’s Combinational Equivalence Checking (CEC).

3.2. Neural Guidance vs. Blind Beam Search

Table 1 reports the synthesis performance of the full CSFM pipeline compared to a blind greedy baseline ($\gamma = 0.0$).

Scalability on Large Manifolds: While blind beam search successfully optimizes small state spaces (achieving identical 70.03% and 64.14% ADP reductions on the `priority` and `ctrl` circuits, respectively), it degenerates into greedy local minima on large scale circuits. For the 57,247 node `div` (divider) benchmark, removing neural guidance traps the discrete search at a 29.53% ADP reduction. Activating the CSFM continuous rollout successfully guides the beam search out of this local minimum, eliminating over 36,000 redundant gates and achieving a 63.66% ADP reduction.

Algorithmic Safety on Flat Topologies: The `dec` (decoder) and `arbiter` circuits feature extreme, structurally flat topologies (e.g., `dec` has a maximum critical path depth of 3 levels). Standard atomic transformations in the action library yield no valid actions where the normalized cost $g(s') < 1.0$. Consequently, the discrete beam search safely terminates at the root state for both guided and blind search baselines, confirming the framework acts as a robust safeguard against destructive area bloat.

Table 1. Multi Objective Synthesis Performance on Unseen EPFL Benchmarks (Balanced $\alpha_{\text{area}} = 0.5$, $\alpha_{\text{delay}} = 0.5$). CSFM Guidance is compared against Blind Greedy Search ($\gamma = 0$). All optimized graphs are formally verified (CEC).

Benchmark Name	Initial Circuit			CSFM Guided (Ours)			Relative Reduction (%)			Blind Search Δ ADP
	Area	Delay	ADP	Area	Delay	ADP	Δ Area	Δ Delay	Δ ADP	
div	57,247	4,372	250,283,884	20,757	4,382	90,957,174	63.74%	-0.23%	63.66%	29.53%
priority	978	250	244,500	568	129	73,272	41.92%	48.40%	70.03%	70.03%
ctrl	174	10	1,740	104	6	624	40.23%	40.00%	64.14%	64.14%
arbiter	11,839	87	1,030,000	11,839	87	1,030,000	0.00%	0.00%	0.00%	0.00%
dec	304	3	912	304	3	912	0.00%	0.00%	0.00%	0.00%

Table 2. Ablation Study on Representation Collapse (Metrics shown for the `div` benchmark).

Configuration	Δ Area	Δ Delay	Δ ADP
Proposed CSFM (Full)	63.74%	-0.23%	63.66%
No Alignment ($\lambda_{\text{align}} = 0$)	63.79%	-0.25%	63.70%
No Var. Reg. ($\lambda_{\text{var}} = 0$)	29.19%	+0.48%	29.53%

Table 3. Continuous Pareto Steering on Unseen Benchmarks.

Circuit	Objective Weight (w)	Δ Area	Δ Delay
div	area heavy ($\alpha_A = 0.9, \alpha_D = 0.1$)	63.82%	-0.46%
	Balanced ($\alpha_A = 0.5, \alpha_D = 0.5$)	63.74%	-0.23%
	delay heavy ($\alpha_A = 0.1, \alpha_D = 0.9$)	28.74%	0.48%
priority	area heavy ($\alpha_A = 0.9, \alpha_D = 0.1$)	43.35%	45.60%
	Balanced ($\alpha_A = 0.5, \alpha_D = 0.5$)	41.92%	48.40%
	delay heavy ($\alpha_A = 0.1, \alpha_D = 0.9$)	50.72%	66.00%

3.3. Ablation on Representation Collapse

We ablate the structural flow regularizations derived in Section 2 to validate their empirical necessity in preventing latent space mode collapse. Results on the `div` benchmark are reported in Table 2.

Removing Variance Regularization ($\lambda_{\text{var}} = 0$) causes total mode collapse. The generated source distribution collapses into a trivial mapping, making the neural heuristic mathematically useless. Consequently, the search degenerates precisely to the blind search baseline (29.53% ADP reduction).

Conversely, removing Directional Alignment ($\lambda_{\text{align}} = 0$) resulted in a 63.70% ADP reduction, performing marginally on par with the full model. This indicates that the combination of Variance Regularization ($\mathcal{L}_{\text{VarReg}}$) and batch wise standard deviation penalization ($\mathcal{L}_{\text{std}}$) is the primary stabilizing mechanism preventing encoder collapse, while explicit alignment acts merely as an auxiliary dynamic during early training epochs.

3.4. Continuous Pareto Multi Objective Steering

To validate that the conditional flow matching model effectively models the continuous Pareto frontier, we evaluated the framework using disparate objective weights on the `div` and `priority` benchmarks (Table 3).

The continuous representations strictly shift the discrete search trajectory according to the provided preference vector. For the `div` circuit, the area heavy configuration ($\alpha_{\text{area}} = 0.9$) maximized area reduction (63.82%) at the expense of delay (-0.46%). Conversely, the delay heavy configuration ($\alpha_{\text{delay}} = 0.9$) successfully pivoted the trajectory, sacrificing massive area reductions (dropping to 28.74%) to achieve a positive delay reduction (+0.48%).

Similarly, for the `priority` benchmark, shifting from an area heavy to a delay heavy preference induced a massive 20.4% absolute leap in delay reduction (from 45.60% to 66.00%). These results empirically confirm that the CSFM model successfully parameterizes the continuous multi objective design space, allowing dynamic, user specified logic synthesis without requiring model retraining.

4. Conclusion and Broader Impacts

In this work, we presented Neuro-Symbolic CSFM, a scalable framework that successfully bridges continuous generative flow matching and discrete logic synthesis. By projecting combinatorial optimization into a regularized latent space, we demonstrated that continuous ODE trajectories can serve as highly effective, steerable heuristics for strict, equivalence preserving beam search. Our empirical evaluations confirmed that explicitly regularizing the variance and batch wise covariance of the latent space is strictly necessary to prevent mode collapse when targeting deterministic logic structures.

Consequently, the proposed framework accurately navigates continuous area delay Pareto tradeoffs and achieves significant structural reductions on massive, industrial scale benchmarks where unguided heuristics degenerate into local minima. By ensuring that all final structural edits are routed through verified atomic transformations, Neuro-Symbolic

CSFM completely eliminates the risk of Boolean logic faults, satisfying the strict exactness constraints of modern hardware compilation. Immediate extensions within the EDA domain include applying continuous flow guidance to sequential synthesis (e.g., register retiming) and standard cell technology mapping.

Broader Implications: Beyond hardware design, the Neuro-Symbolic CSFM paradigm offers a generalized solution to domains where continuous generative relaxations fail due to strict discrete constraints. In software compiler superoptimization e.g., LLVM IR etc. (Cummins et al., 2021; Schkufza et al., 2012), where continuous language models frequently hallucinate invalid syntax, our framework could guarantee exact semantic preservation. Similarly, in quantum circuit synthesis, this architecture naturally maps the continuous representation of target unitary matrices to discrete, fault tolerant gate sets e.g., Clifford+T etc. (Giles & Selinger, 2013; Dawson & Nielsen, 2005). Finally, in de novo molecular and crystal graph generation, where continuous diffusion models frequently violate valency physics, our flow guided discrete assembly (Hoogeboom et al., 2022) can guarantee 100% physically synthesizable structures (Vignac et al., 2023; You et al., 2019), establishing a robust new bridge between continuous representation learning and strict combinatorial search.

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